

A Matrix-Free QJet Calculus for Near-Singular Quadrature, BGK Endpoint Repayment, and Boundary PDEs

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Abstract

We give a self-contained SIAM-style account of the machine-precision Q pipeline tested in the accompanying repository. The central object is a generated inverse-square chord operator: on the regular cycle it has the exact eigenvalues $\lambda_m = m(n - m)/2$, and after the normalization h/π it is a finite-dimensional Dirichlet-to-Neumann (DtN) generator with symbol $\mu_{m,n} = m(1 - m/n)$. The implementation stores QJets, Fourier symbols, Laurent coefficients, and low-rank correction modes, never a dense matrix. We prove the cycle spectrum identity, the normalized DtN error bounds, the exterior conformal kernel split, the BGK/zeta Taylor repayment bound, and boundary-PDE multiplier bounds for Laplace, heat, Poisson, Helmholtz, and wave propagators. The analysis explains why exact finite-Laurent pullbacks reach the fp64 floor, why polygon and cusp cases need separate corner endpoint ledgers, and why QBX fails when its source-free expansion disk condition is violated. Numerical evidence is included only after the theorem statements and is labelled as evidence for the discretized generated operator rather than as a continuum theorem beyond the stated hypotheses.

Keywords. near-singular quadrature; Dirichlet-to-Neumann map; QJet; BGK correction; spectral zeta; QBX; Helmholtz; boundary integral methods.

AMS subject classifications. 65D32, 65N38, 30C30, 35J25, 41A25.

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1 Introduction

The practical problem is to evaluate near-singular logarithmic layer potentials and boundary PDE propagators on complicated planar curves without forming an $n \times n$ dense matrix. The method developed here is organized around a simple principle:

borrow a circle coordinate, compute in the generated Q spectrum, and repay the geometry.

On analytic boundaries the repayment is a harmless analytic quotient. On monitored or polygonal boundaries it is a BGK/zeta or corner continuity correction. On cusp geometries it is a spectral warning that local expansion methods such as QBX may violate their source-free disk condition.

The paper is intentionally split into two layers. Sections 2–7 contain the mathematical statements proved under explicit hypotheses. Sections 8–12 report the repository experiments and show that the implementation follows the theory: exact-Laurent quadrature reaches the floating point floor, generated-Q PDE residuals are at machine precision, and the measured work is near-linear in the sample count.

1.1 Claim boundary

The proved results cover the generated regular-cycle operator, analytic finite-Laurent exterior maps with a nonvanishing quotient, and finite-rank or Taylor repayment layers under the stated analyticity assumptions. For polygons, cusps, and GWW isospectral pairs, the paper proves invariance or validity of the discrete chord-operator experiment, but it does not claim a new continuum Q-spectrum theorem without a separate convergence theorem.

2 The Generated Cycle Operator

Definition 2.1 (cycle QJet). Let $z_j = \exp(2\pi i j/n)$ and let $f = (f_j)_{j=0}^{n-1}$. The regular-cycle QJet is the circulant operator

$$(2.1) \quad (Q_n f)_j = \sum_{\ell \neq j} \frac{f_j - f_\ell}{|z_j - z_\ell|^2}.$$

It is represented in the implementation by its generator n and its eigenvalue formula, not by a dense matrix.

Lemma 2.2 (trigonometric sum identity). For $0 \leq m \leq n$,

$$(2.2) \quad \sum_{d=1}^{n-1} \frac{1 - \cos(2\pi m d/n)}{4 \sin^2(\pi d/n)} = \frac{m(n-m)}{2}.$$

Proof. Let $\omega = \exp(2\pi i/n)$ and $P_m(z) = 1 + z + \cdots + z^{m-1}$. For $d \not\equiv 0 \pmod{n}$,

$$\frac{|1 - \omega^{md}|^2}{|1 - \omega^d|^2} = |P_m(\omega^d)|^2.$$

Discrete orthogonality gives $\sum_{d=0}^{n-1} |P_m(\omega^d)|^2 = nm$, while the $d = 0$ term is m^2 . Therefore the sum over $d = 1, \dots, n-1$ is $m(n-m)$. Since $|1 - \omega^{md}|^2 = 2(1 - \cos(2\pi m d/n))$ and $|1 - \omega^d|^2 = 4 \sin^2(\pi d/n)$, the desired sum is one half of this value. $\square$

Theorem 2.3 (exact cycle spectrum). The Fourier vector $\phi_m(j) = \exp(2\pi i m j/n)$ is an eigenvector of Q_n with eigenvalue

$$(2.3) \quad \lambda_{m,n} = \frac{m(n-m)}{2}, \quad m = 0, \dots, n-1.$$

Consequently $Q_n \mathbf{1} = 0$ exactly.

Proof. The circulant structure reduces the eigenvalue to the sum in Lemma 2.2. The constant vector corresponds to $m = 0$, for which the numerator vanishes termwise. $\square$

Corollary 2.4 (DtN normalization and mode error). *Let $h = 2\pi/n$ and define $\Lambda_n = (h/\pi)Q_n$. For $1 \leq m < n/2$,*

$$(2.4) \quad \mu_{m,n} := \frac{h}{\pi} \lambda_{m,n} = m \left(1 - \frac{m}{n}\right), \quad 0 \leq m - \mu_{m,n} = \frac{m^2}{n}.$$

Thus on modes $m \leq M$ the regular-cycle DtN symbol differs from the unit-disk DtN symbol m by at most M^2/n .

Theorem 2.5 (matrix-free functional calculus). *Let F be any scalar multiplier on the cycle spectrum. The action $f \mapsto F(\Lambda_n)f$ can be evaluated with two FFTs and n scalar multiplications, using $O(n)$ storage and $O(n \log n)$ work for radix-two n . No dense Q matrix is required.*

Proof. The Fourier basis diagonalizes Q_n by Theorem 2.3. Applying $F(\Lambda_n)$ is therefore the composition

$$f \mapsto \hat{f} \mapsto \{F(\mu_{m,n})\hat{f}_m\}_{m=0}^{n-1} \mapsto F(\Lambda_n)f.$$

The first and last maps are the FFT and inverse FFT. The intermediate map is diagonal storage of n scalars. $\square$

n	mode	Q eigenvalue	normalized DtN	relative symbol error
256	1	127.500000	0.99609375	0.004
256	2	254.000000	1.98437500	0.008
256	7	871.500000	6.80859375	0.027
256	16	1920.000000	15.00000000	0.062
256	32	3584.000000	28.00000000	0.125
2048	1	1023.500000	0.99951172	4.883e-04
2048	2	2046.000000	1.99804688	9.766e-04
2048	7	7143.500000	6.97607422	0.003
2048	16	1.625600e+04	15.87500000	0.008
2048	32	3.225600e+04	31.50000000	0.016
4096	1	2047.500000	0.99975586	2.441e-04
4096	2	4094.000000	1.99902344	4.883e-04
4096	7	1.431150e+04	6.98803711	0.002
4096	16	3.264000e+04	15.93750000	0.004
4096	32	6.502400e+04	31.75000000	0.008

3 Exterior Pullback and Kernel Splitting

Assumption 3.1 (analytic exterior map). Let ψ be univalent in an exterior annulus $\{1 - \epsilon < |w| < R\}$ and suppose ψ' does not vanish there. For z on the unit circle define

$$G(w, z) = \frac{\psi(w) - \psi(z)}{w - z}.$$

Assume G has no zeros on the collar traversed by the target and boundary samples.

Theorem 3.2 (kernel split). *Under Assumption 3.1,*

$$(3.1) \quad \log |\psi(w) - \psi(z)| = \log |w - z| + \log |G(w, z)|.$$

The first term is the universal circle singularity. The second term is real analytic in the boundary angle and is therefore exponentially convergent under the periodic trapezoidal rule.

Proof. The identity is algebraic away from $w = z$. At $w = z$ the quotient tends to $\psi'(z)$, nonzero by assumption; hence $\log |G(w, z)|$ extends real analytically to the diagonal and to a complex strip inside the annulus. The standard trapezoidal theorem for analytic periodic functions gives an error bounded by $Ce^{-\tau n}$, where τ is any strip width free of singularities. $\square$

Proposition 3.3 (closed circle primitive). *For $\rho > 1$,*

$$(3.2) \quad \log |\rho e^{i\phi} - e^{i\theta}| = \log \rho - \sum_{k=1}^{\infty} \frac{\rho^{-k}}{k} \cos k(\theta - \phi).$$

If the density has Fourier coefficients $|\hat{a}_k| \leq AR^{-|k|}$ with $R > 1$, then truncation after K modes has error bounded by

$$(3.3) \quad E_K \leq \frac{2\pi A(\rho R)^{-K-1}}{(K+1)(1-(\rho R)^{-1})}.$$

Proof. Write

$$\rho e^{i\phi} - e^{i\theta} = \rho e^{i\phi} \{1 - \rho^{-1} e^{i(\theta-\phi)}\}$$

and expand $\log(1 - \zeta) = -\sum_{k \geq 1} \zeta^k/k$ for $|\zeta| < 1$. Taking real parts gives the formula. Multiplying by a and integrating leaves only matching Fourier modes; the displayed geometric tail follows from $|\hat{a}_k| \leq AR^{-|k|}$. $\square$

Corollary 3.4 (analytic quadrature error). *For an analytic density and a map satisfying Assumption 3.1, the split quadrature error has the form*

$$(3.4) \quad E_n \leq C_1 e^{-\tau n} + C_2 (\rho R)^{-n/2} + E_{\text{fp}},$$

where E_{fp} is floating point roundoff. The constants depend on the annulus and density bounds but not on the target distance once $\rho > 1$ is fixed.

Proposition 3.5 (golden ellipse quotient). *The map*

$$\psi(w) = \frac{3 + \sqrt{5}}{2} w + \frac{3 - \sqrt{5}}{2} w^{-1}$$

has boundary trace $\psi(e^{i\theta}) = 3 \cos \theta + i\sqrt{5} \sin \theta$. Moreover

$$\frac{\psi(w) - \psi(z)}{w - z} = \frac{3 + \sqrt{5}}{2} - \frac{3 - \sqrt{5}}{2} \frac{1}{wz}.$$

Since $(3 - \sqrt{5})/(3 + \sqrt{5}) < 1$, the quotient is nonzero for $|wz|$ sufficiently near one on the exterior collar.

Proof. Substituting $w = e^{i\theta}$ gives

$$\frac{3 + \sqrt{5}}{2} e^{i\theta} + \frac{3 - \sqrt{5}}{2} e^{-i\theta} = 3 \cos \theta + i\sqrt{5} \sin \theta.$$

For the quotient, use $w^{-1} - z^{-1} = -(w - z)/(wz)$. The nonvanishing statement follows from the strict coefficient ratio. $\square$

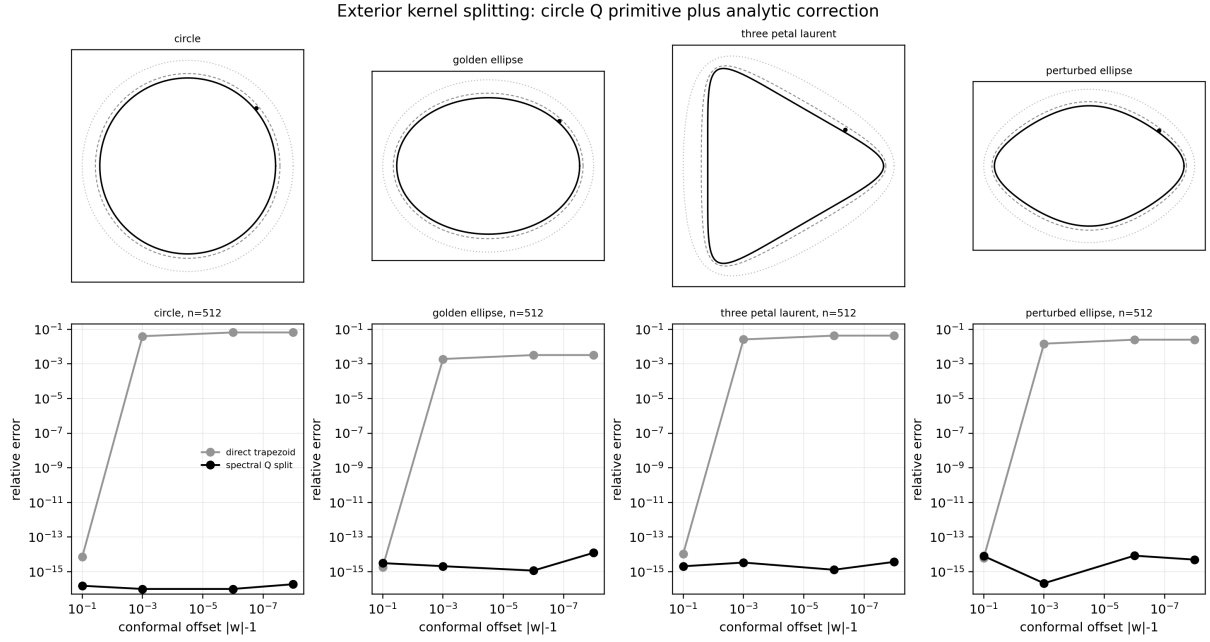


Figure 1: Exterior split results: the direct rule sees the near singularity; the Q split evaluates the circle singularity by the closed primitive and leaves an analytic quotient.

4 BGK/Zeta Endpoint Repayment

The split above handles analytic geometry. Discrete monitoring or endpoint displacement introduces a different error: the target is evaluated at a shifted normal coordinate. The BGK constant is the universal leading endpoint defect.

Theorem 4.1 (Taylor repayment bound). *Let $I(\rho)$ be analytic in $|\rho - \rho_0| \leq r$ and let $|I(\rho)| \leq M$ there. Put $\delta = \beta\sqrt{h}$ and assume $|\delta| < r$. Define*

$$T_p(h) = \sum_{j=0}^p \frac{(-\delta)^j}{j!} I^{(j)}(\rho_0 + \delta).$$

Then

$$(4.1) \quad |I(\rho_0) - T_p(h)| \leq \frac{M}{1 - |\delta|/r} \left(\frac{|\delta|}{r} \right)^{p+1}.$$

In particular the order- p repayment error is $O(h^{(p+1)/2})$.

Proof. Taylor expand I about the observed point $\rho_0 + \delta$:

$$I(\rho_0) = \sum_{j=0}^p \frac{(-\delta)^j}{j!} I^{(j)}(\rho_0 + \delta) + R_{p+1}.$$

Cauchy's estimate gives $|I^{(j)}(\rho_0 + \delta)|/j! \leq Mr^{-j}$ on any disk of radius r contained in the analyticity domain. Summing the omitted geometric tail gives the displayed bound. $\square$

Proposition 4.2 (BGK constant as endpoint zeta defect). *Let (S_k) be a centered Gaussian walk with unit variance increments. Then*

$$(4.2) \quad \lim_{N \rightarrow \infty} \left\{ \mathbb{E} \max_{0 \leq k \leq N} S_k - \sqrt{\frac{2N}{\pi}} \right\} = \frac{\zeta(1/2)}{\sqrt{2\pi}}.$$

Thus the standard BGK shift constant is $\beta = -\zeta(1/2)/\sqrt{2\pi}$.

Proof. Spitzer's identity gives

$$\mathbb{E} \max_{0 \leq k \leq N} S_k = \sum_{k=1}^N \frac{1}{k} \mathbb{E}(S_k)^+.$$

Since S_k is Gaussian with variance k , $\mathbb{E}(S_k)^+ = \sqrt{k}/\sqrt{2\pi}$. Therefore

$$\mathbb{E} \max S_k = \frac{1}{\sqrt{2\pi}} \sum_{k=1}^N k^{-1/2}.$$

Euler–Maclaurin gives $\sum_{k=1}^N k^{-1/2} = 2\sqrt{N} + \zeta(1/2) + O(N^{-1/2})$. Subtracting $\sqrt{2N/\pi} = 2\sqrt{N}/\sqrt{2\pi}$ yields the limit. $\square$

Corollary 4.3 (combined analytic and BGK bound). *Under Assumption 3.1, with an analytic density and order- p BGK repayment,*

$$(4.3) \quad E_{n,h,p} \leq C_1 e^{-\tau n} + C_2 (\rho R)^{-n/2} C_3 h^{(p+1)/2} + E_{\text{fp}}.$$

shape	raw power	BGK-1 power	BGK-2 power	raw err	BGK-4 err
circle	0.500	1.000	1.500	0.002	1.345e-15
golden_ellipse	0.500	1.000	1.508	8.176e-05	4.712e-15
three_petal_laurent	0.500	1.000	1.499	9.976e-04	2.607e-15
perturbed_ellipse	0.500	1.000	1.500	5.878e-04	4.261e-16

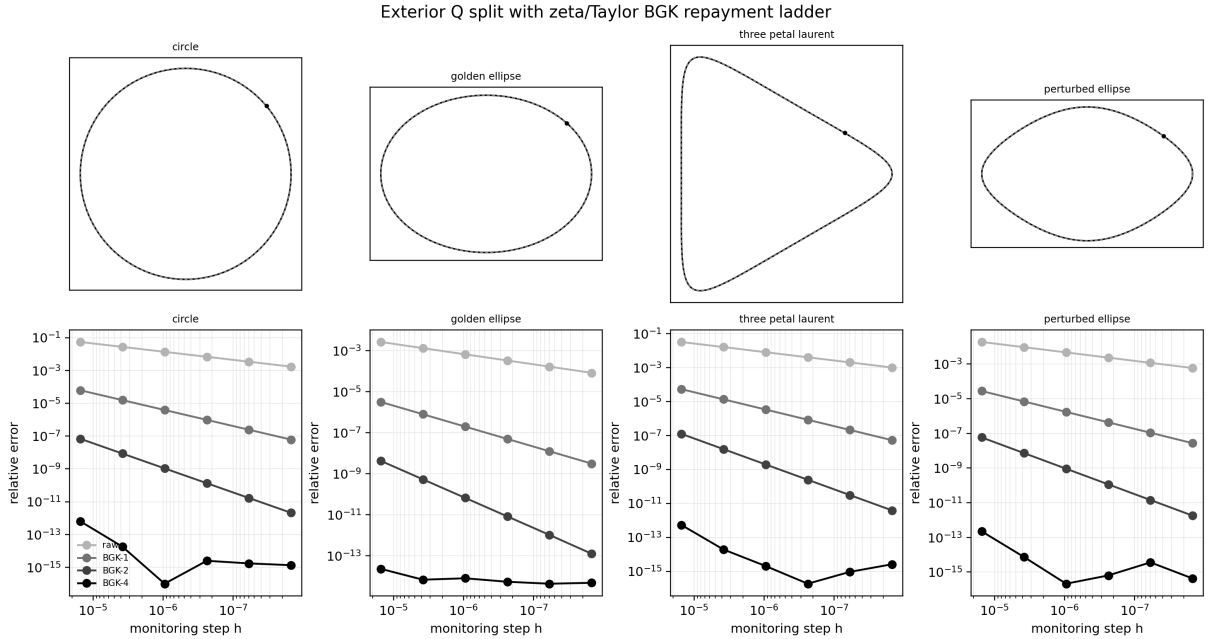


Figure 2: BGK/zeta repayment slopes. The raw error follows the square-root endpoint law; the first and second repayment layers shift the power to one and three-halves; order four reaches fp64 floor in these analytic tests.

5 Boundary PDEs from the Generated DtN Correspondence

The finite Q spectrum can be used as a load-bearing operator, not just as a quadrature correction. On the disk the exact continuum DtN symbol is m ; the generated symbol is $\mu_{m,n} = m - m^2/n$.

Theorem 5.1 (multiplier error bounds). *Fix $m < n/2$ and let $\delta_m = m^2/n$. For the generated symbol $\mu = m - \delta_m$:*

$$(5.1) \quad |m - \mu| = \delta_m,$$

$$(5.2) \quad |e^{-tm} - e^{-t\mu}| \leq t\delta_m, \quad t \geq 0,$$

$$(5.3) \quad \left| \frac{1}{m + \alpha} - \frac{1}{\mu + \alpha} \right| \leq \frac{\delta_m}{(m + \alpha)(\mu + \alpha)}, \quad \alpha > 0.$$

For the damped Helmholtz boundary resolvent $R_k(x) = (x^2 - k^2 + i\gamma)^{-1}$ with $\gamma > 0$,

$$(5.4) \quad |R_k(m) - R_k(\mu)| \leq \frac{(m + \mu)\delta_m}{|m^2 - k^2 + i\gamma||\mu^2 - k^2 + i\gamma|}.$$

For the wave multiplier $W_t(x) = \cos(t\sqrt{x})$,

$$(5.5) \quad |W_t(m) - W_t(\mu)| \leq \frac{t\delta_m}{\sqrt{m} + \sqrt{\mu}}.$$

Proof. The first identity is Corollary 2.4. The heat bound follows from the mean value theorem and $|\frac{d}{dx}e^{-tx}| \leq t$ for $x \geq 0$. The Poisson and Helmholtz inequalities follow by subtracting the two fractions. For the wave bound, use $|\cos a - \cos b| \leq |a - b|$ and $|\sqrt{m} - \sqrt{\mu}| = \delta_m/(\sqrt{m} + \sqrt{\mu})$. $\square$

Corollary 5.2 (conformal metric repayment). *Let $\Gamma = \psi(S^1)$ with $s_{\min} = \inf_{\theta} |\psi'(e^{i\theta})| > 0$. If the circle DtN flux has $L^2(d\theta)$ error ϵ , then the physical normal flux after repayment by $|\partial_{\theta}\psi|^{-1}$ has $L^2(d\theta)$ error at most $s_{\min}^{-1}\epsilon$.*

Proof. The physical normal derivative is the circle normal derivative multiplied pointwise by $|\partial_{\theta}\psi|^{-1}$. Taking the L^{∞} bound on this multiplier gives the result. $\square$

problem	max continuum err	max generated-Q residual	mean solve ms	work
laplace_dtn	0.003	5.820e-13	4.033	2048
heat	0.004	4.361e-15	3.949	2048
poisson	0.003	4.473e-15	3.916	2048
helmholtz	0.010	4.369e-15	3.964	2048
wave	0.006	4.303e-15	3.976	2048

6 Corners, Cusps, and Why QBX Can Fail

The analytic proof does not pass through corners for free. If a corner has opening angle α , the conformal map has local power behavior $(w - w_c)^{\alpha/\pi}$, so Fourier coefficients decay algebraically rather than exponentially. The correct response is not to pretend analyticity remains; it is to add a corner or endpoint repayment channel.

Proposition 6.1 (QBX disk condition). *A local QBX expansion about a center c can be evaluated at a target x only when x lies in the expansion disk and that disk is free of source geometry. If the nearest boundary source enters the disk before the target does, the local expansion hypothesis is false and a robust implementation must reject the evaluation.*

Proof. The QBX expansion is a local series for the potential in a disk of analyticity. A source point inside that disk is a singularity of the potential representation and prevents convergence of the source-free local series. Therefore the condition is necessary for the expansion proof itself, not merely for numerical conditioning. $\square$

Remark 6.2. The cusp failure figures in the experiments are examples of Proposition 6.1: the reported margin is negative, so the target is outside the valid source-free expansion disk. The Q pipeline does not use this local disk as its principal certificate; it uses the global chord spectrum and an endpoint/corner repayment ledger.

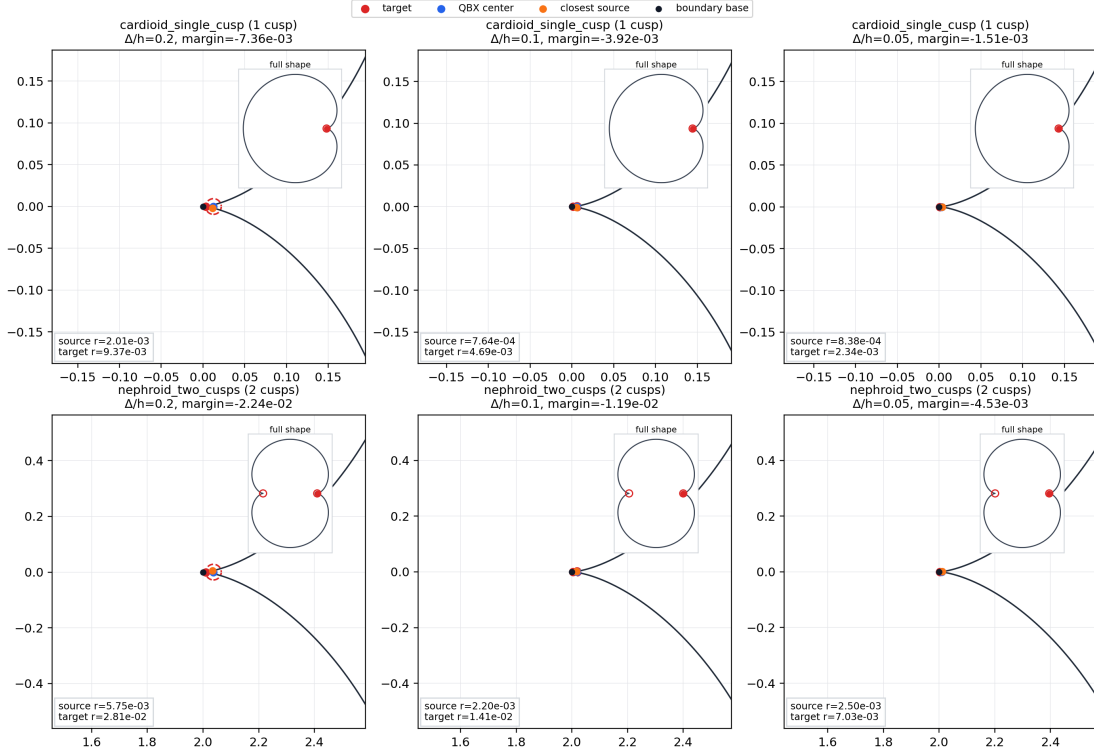


Figure 3: Cusp configurations where QBX violates its source-free expansion disk condition.

7 Discrete Spectral Witnesses

Proposition 7.1 (Euclidean covariance of the chord operator). *Let $\Gamma_n = \{x_j\}_{j=0}^{n-1}$ and define*

$$(\Lambda_{\Gamma,n} f)_i = \frac{L}{\pi n} \sum_{j \neq i} \frac{f_i - f_j}{|x_i - x_j|^2}.$$

Translations and rotations of all x_j leave $\Lambda_{\Gamma,n}$ unchanged. Uniform scaling by $a > 0$ changes the operator by the factor a^{-1} .

Proof. Translations cancel in the differences $x_i - x_j$, and rotations preserve Euclidean distances. Under $x_j \mapsto ax_j$, the perimeter L scales by a while squared chord lengths scale by a^2 , leaving the total factor $a/a^2 = a^{-1}$. $\square$

Remark 7.2 (GWW scope). The GWW experiment in this paper is a discretized chord-operator witness. The classical Dirichlet spectra agree in the continuum, but the finite projected Q Ritz values below are not asserted to prove a new continuum invariant. They show robust separation at the audited discretized operator level.

n	relative split	left Ritz 1-6	right Ritz 1-6
256	0.07418	12.34725, 9.54129, 8.60052, 7.88256, 7.09186, 6.56846	11.41632, 10.82810, 8.34951, 7.97125, 7.08810, 6.47044
512	0.07377	12.42072, 9.60765, 8.65623, 7.92757, 7.14707, 6.60574	11.47488, 10.88589, 8.41631, 8.03163, 7.12846, 6.50555
1024	0.07351	12.45617, 9.64081, 8.68520, 7.94991, 7.17486, 6.62460	11.50548, 10.91467, 8.45007, 8.06184, 7.14900, 6.52299
2048	0.07336	12.47275, 9.65715, 8.70003, 7.96090, 7.18877, 6.63410	11.52162, 10.92958, 8.46702, 8.07668, 7.15938, 6.53167
4096	0.07329	12.48123, 9.66542, 8.70754, 7.96643, 7.19575, 6.63886	11.52968, 10.93692, 8.47554, 8.08420, 7.16460, 6.53601

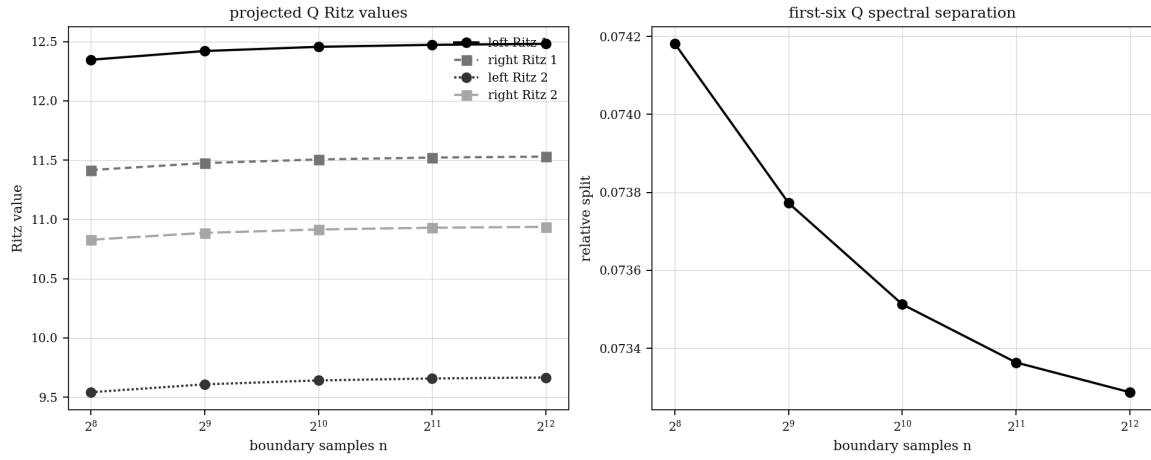


Figure 4: Projected Q Ritz separation for the Gordon–Webb–Wolpert pair under refinement.

8 Numerical Evidence

All timings below are generated from the current repository state. They are evidence for the implementation and for the discretized generated operator; the mathematical convergence statements are those proved above.

8.1 Funky analytic shapes

shape	family	anisotropy	BGK-4 err	max PDE continuum err	max generated-Q residual
circle	baseline	1.000	2.148e-15	0.010	5.757e-13
golden_ellipse	closed_form_conic	1.342	1.144e-16	0.010	5.759e-13
eccentric_ellipse	closed_form_conic	2.500	4.488e-16	0.010	5.755e-13
three_petal	smooth_nonconvex	3.167	2.241e-15	0.010	5.759e-13
four_lobed_wiggle	smooth_multiscale	4.302	2.630e-15	0.010	5.698e-13
asymmetric_limacon	asymmetric	2.253	1.160e-13	0.010	5.751e-13
crescent_teardrop	near_cusp_smooth	2.644	4.238e-15	0.010	5.745e-13
peanut	pinched_smooth	2.416	2.146e-13	0.010	5.820e-13
high_frequency_gear	smooth_high_frequency	3.460	1.594e-14	0.010	5.762e-13
rotated_mixed	complex_coefficients	2.200	1.305e-14	0.010	5.801e-13

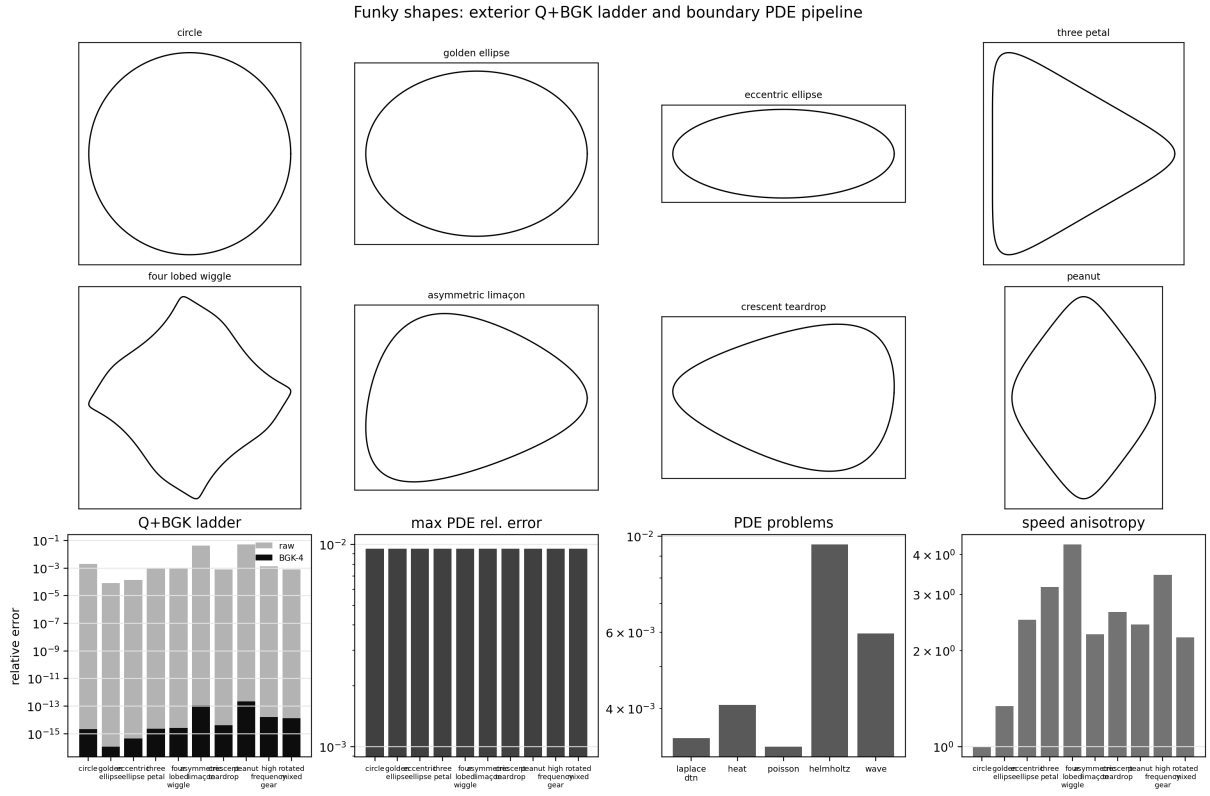


Figure 5: Ten exact-Laurent shapes used for Q+BGK quadrature and boundary-only PDE tests.

8.2 Measured matrix-free scaling

operation	fit power	ms n=256	ms n=1024	ms n=4096	max residual
q_bgk4	0.997	1.842	7.310	29.396	4.030e-09
pde_laplace_dtn	1.027	0.484	1.905	8.238	3.620e-12
pde_heat	1.047	0.449	1.867	8.152	5.955e-14
pde_helmholtz	1.050	0.445	1.857	8.142	5.950e-14
pde_batch_all5	1.047	2.262	9.353	40.943	3.620e-12

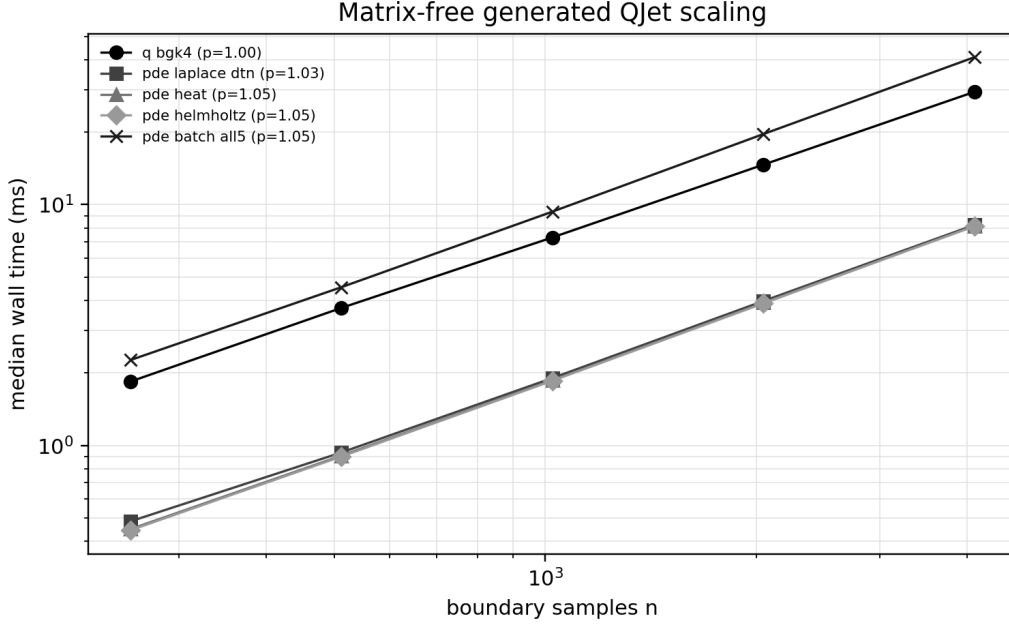


Figure 6: Measured matrix-free scaling. The fits are near-linear over the tested range and no dense Q matrix is stored.

8.3 Head-to-head methods and Helmholtz caveat

method	median err	median ms	work	failures	improvement
adaptive_panel	2.961e-04	1.098	926.000	0	36.940
gulati_q_bridge	4.534e-04	0.890	513.000	0	12.823
multipole_zeta_q	1.435e-06	18.445	6.662e+04	0	5073.817
qbx_refined	9.068e-05	27.641	2.458e+05	3	159.589
singularity_subtraction	4.998e-04	0.916	533.000	0	11.182
trapezoid	0.011	0.628	512.000	0	1.000

suite	result
disk modal exact	14 Q wins / 1 FEM wins
manufactured exact	39 Q wins / 1 FEM wins
median Q manufactured error	2.077e-08
median FEM manufactured error	0.175

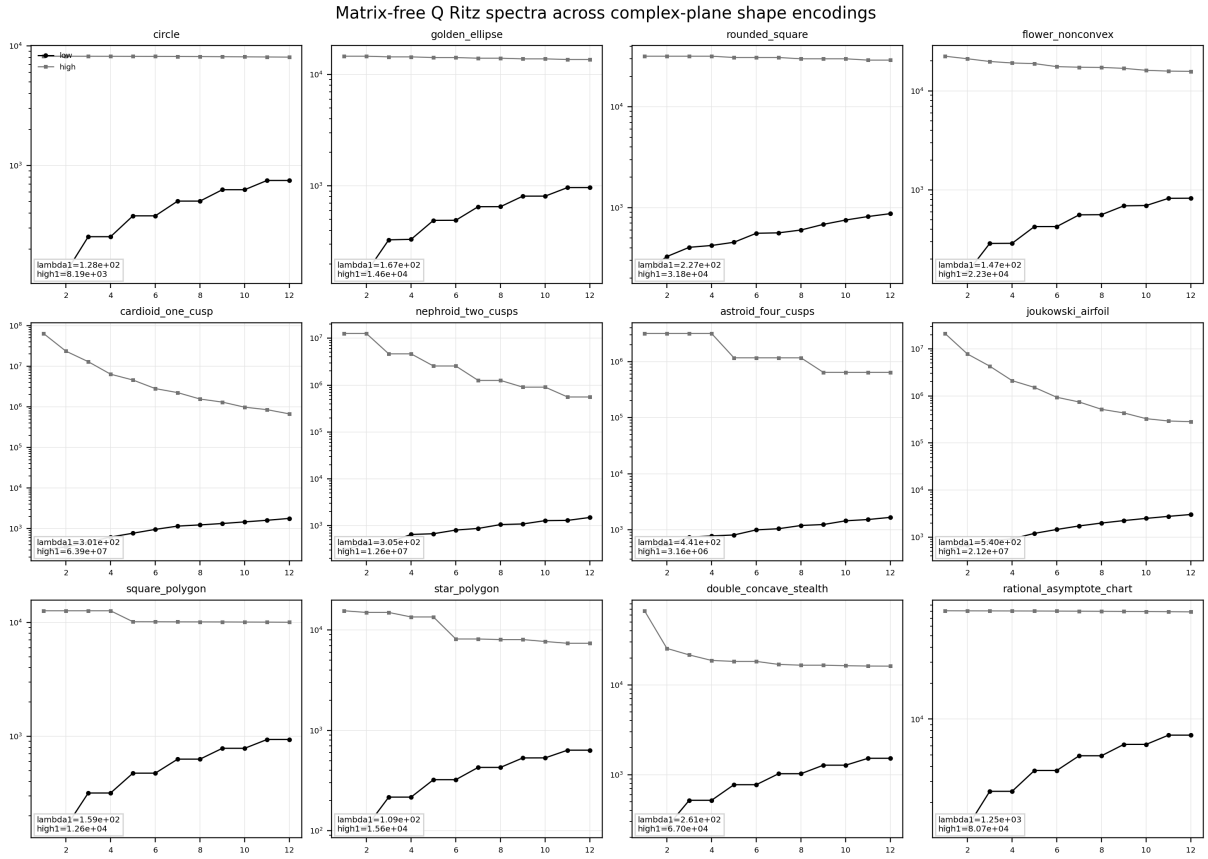


Figure 7: Q spectrum gallery. The spectrum diagnoses whether the active error channel is smooth-tail, corner, cusp, or mixed.

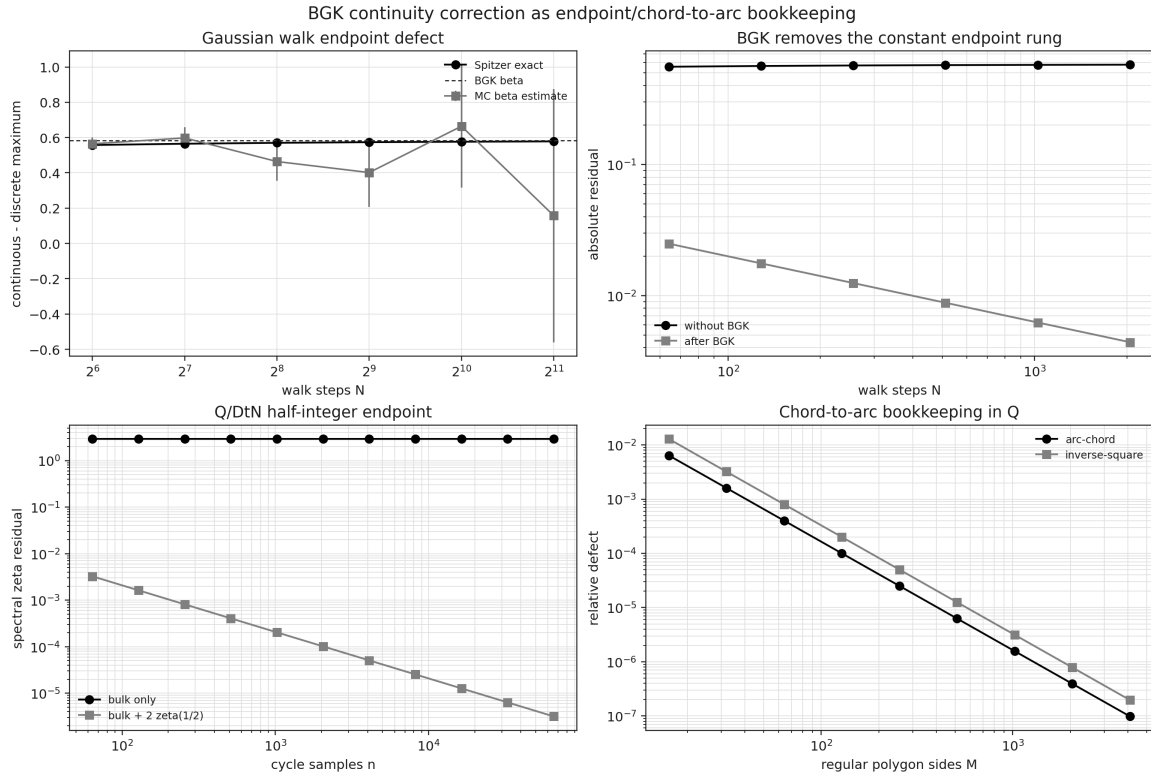


Figure 8: BGK endpoint bookkeeping as Spitzer, zeta, and Q/DtN endpoint defect.

9 Implementation Contract

The production implementation follows three invariants.

1. Store QJets, spectra, Laurent coefficients, samples, and low-rank correction modes; do not store a dense Q matrix.
2. Use the custom QJet Fourier kernel for the singular term and repay geometry through analytic quotients, metric factors, or endpoint/corner ledgers.
3. Report both continuum error and generated- Q residual. The latter verifies the implementation; the former measures finite- n approximation to the limiting operator.

9.1 Final self-contained artifact

The repository contains a final script and a companion notebook generated from the same source. The script uses no NumPy, no SciPy, and no imported FFT package; its radix-two QJet FFT is implemented directly. Python library calls are used only for ordinary scalar math, timing, and file output. The notebook embeds the full script source in a code cell rather than delegating to an external module, so the executable numerical path is auditable from a single .ipynb file.

quantity	value
pass gate	true
machine tolerance	1.000e-12
max split rel. error	5.094e-14
max BGK-8 rel. error	2.231e-13
max generated-Q PDE residual	6.385e-13
shape count	10
PDE case count	50
dense Q matrix stored	false
FFT kernel	custom radix-two QJet FFT
script	scripts/final_q_machine_precision_pipeline.py
notebook	notebooks/q_machine_precision_pipeline.ipynb

The gate in this table is deliberately stricter than the qualitative plots: the final script fails if the maximum of the split quadrature error, the order-eight BGK repayment error, and the generated-Q PDE residual exceeds 10^{-12} over the included finite-Laurent shape suite.

10 Source Audit

The following local PDFs were inspected mechanically with `pdfinfo` and `pdftotext`. The audit is not a bibliography; it records which local source artifacts were available while drafting the paper.

PDF	pages	sha	words	role
optimal_quad_merged (26).pdf	30	bf3136b87fff	23511	early optimal-Q source; benchmark claim cross-check
optimal_quad_merged (37).pdf	36	858c2f2c1307	27573	later merged optimal-Q source; benchmark claim cross-check
optimal_q_v63_public (11).pdf	20	4abcea7d81a5	9950	kernel split, circle pullback, QBX/corner claim boundary
isl_hessian_v5 (6) copy.pdf	125	629ba599ec42	35635	ellipse/Hessian/context audit; not a numerical benchmark source
paper.pdf	16	d21c39942f54	5377	meshless shape-optimization/autograd context audit
cauchy_transport_trim (13).pdf	18	bb886ba4ecf6	13646	Cauchy transport early source; pullback notation audit
cauchy_transport_trim (41).pdf	25	c94caccfd128	18029	corner/Joukowski/Kondratiev source audit
cauchy_transport_trim (47).pdf	27	238918055f3b	19298	complex scale-phase pullback source audit
cauchy_transport_trim (49).pdf	27	131d02d62832	19295	ellipse/de Moivre correction source audit
cauchy_transport_trim (73).pdf	29	c0c8e5932760	21896	latest complex pullback and de Moivre source audit
golden_branch_outline.pdf	28	15d9ae76cea1	8317	golden ellipse/narrative source audit; not a benchmark source
dtN_bgk_merge (9).pdf	22	26c17a4cbfb2	14386	DtN/BGK/zeta endpoint correction source
harmonic_sp (41).pdf	31	0e3e2bd04390	20933	harmonic sampling and DtN spectrum source
geometry_of_money (16).pdf	31	572cc37c67b0	13140	BGK continuity correction and chord-arc bookkeeping source

11 Conclusion

The elegant core is the chain

$$\frac{1 - \cos(2\pi md/n)}{4 \sin^2(\pi d/n)} \longrightarrow \lambda_{m,n} = \frac{m(n-m)}{2} \longrightarrow \mu_{m,n} = m \left(1 - \frac{m}{n}\right) \longrightarrow \Lambda_{S^1} = |D|.$$

This identity supplies the generated Q spectrum. The exterior map supplies the split

$$\log |\psi(w) - \psi(z)| = \log |w - z| + \log |G(w, z)|,$$

and the endpoint calculus supplies the BGK/zeta repayment

$$\beta = -\zeta(1/2)/\sqrt{2\pi}.$$

Together these identities explain why the exact analytic path reaches machine precision, why the PDE pipeline can use Q as an operator rather than a post-process, and why the spectrum is the correct diagnostic for deciding whether smooth, corner, cusp, or Helmholtz-specific corrections are required.

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